

essentially the same, the improvements in the drivers and the PVR software seem to have done the trick. All the TV-Tuners were decent in this respect, though we would pick the Hauppauge, the Leadtek WinFast duo and the Pinnacle PCTV Stereo, given a choice. The Tech-Com duo was once again behind the lot, though we can safely attribute this to their poor PVR.

EXTERNAL TV TUNERS

For those who happen to own a laptop (and there are quite a few), an internal TV Tuner is not an option. Some find opening a PC, a daunting task, or some motherboards may not have a free PCI slot to accommodate an internal PCI TV Tuner card. In such cases, an external TV Tuner is the only viable solution. There are certain advantages that external TV Tuners have over internal ones, such as its better image quality, thanks to the reduction of interference from internal PC components. External TV Tuners are further subdivided into two categories: External USB and External Set Top TV Tuners.

External USB TV Tuners

External USB TV-Tuners can do everything an internal TV Tuner can do. They connect to the PC via the USB 2.0 interface, which sports an ideal data throughput of 480 Mbps. The bandwidth of USB 1.1 (11 Mbps) is insufficient for the data passing between the TV-Tuner and the PC. External TV-Tuners come at a premium, and are bundled with a lot of cables and accessories.

Features

FM Radio

FM radio in an external USB TV Tuner was something we'd never seen before. This time, we actually received two such products: the Compro VideoMate Action Pro and the Hauppauge WinTV-USB-FM.

Hardware Encoder And TV standards

Like FM radio, we had not anticipated a TV-Tuner would boast of a hardware MPEG-2 encoder, but the Leadtek Winfast PalmTop TV has this feature. As already explained, this

Jargon Buster

PVR: A Personal Video Recorder is a software that allows the TV-Tuner to record video content in real-time.

Timeshifting: A feature of the PVR that allows you to pause, replay, or skip a portion of live TV programme.

MPEG: The Moving Picture Experts Group develops standards for digital audio and video compression. There are several versions of this standard, including MPEG-1, MPEG-2, and MPEG-4.

MPEG-1: This standard was designed to code progressive video at a transmission rate of 1.5 Mbps. It is used in Video CDs, but is giving way to the new MPEG-2 standard.

MPEG-2: This standard was designed to code transmission rates above 4 Mbps, and is used in DVD and digital broadcast TV. It is superior to MPEG-1 in terms of picture quality and sound clarity.

MPEG-4: This is a relatively new standard, and is aimed at converging streaming media from different sources. Speech and video synthesis, fractal geometry, and an AI approach to reconstruct images with better, yet lossless, compression.

NTSC: The National Television Standards Committee developed the protocol for colour television broadcast transmission and reception in the US. NTSC signals have been used in the US and Japan since 1954, and have hardly been altered since their inception—except for the addition of new parameters for colour signals like Hue and Tint control. NTSC signals are interlaced, and an NTSC TV image has 525 horizontal lines per frame. Every other line is dropped, so it takes two screen scans to complete one image. 29.97 frames are scanned every second.

PAL: Phase Alternation Line is a standard that was developed in Germany, and is used in India and Europe. In the PAL standard, the horizontal image has 625 horizontal lines per frame. 25 frames are scanned every second. A slight colour variation is seen between the PAL and NTSC standards.

SECAM: *Système Électronique Couleur Avec Mémoire*. SECAM is prevalent in some parts of Europe, mainly France, Russia, and countries of the former Soviet Union, where it was adopted in 1967. In the SECAM system, the luminance information is transmitted in the usual manner, and the chrominance signals is interleaved with it. Colour information is reproduced with minimal error.

NICAM stereo: Near Instantaneous Companded Audio Multiplex stereo was developed by the BBC Research Centre in the early 1980s. It was first transmitted with the PAL colour broadcasting system in Britain. This technology improves on the sound quality of the transmitted TV signal.

offloads the CPU from the encoding task. All the TV-Tuners except for the Leadtek Winfast and Pinnacle PCTV USB2 (which support only PAL) support both PAL and NTSC. This is not much of a concern, as NTSC is non-existent in India.

I/O Ports

Being the most popular of I/O ports for video, S-Video is present across the category. The exceptions are the Compro VideoMate U890 and U900, and the Tech-Com SSD-TV-812Plus. A Composite port is present only in the Compro VideoMate Action Pro and the two Pinnacle PCTVs.

The Remote Control

It's hard to understand why, but the Hauppauge TV-Tuners do not have a remote. Recording and timeshifting are features common to all the TV-Tuners, but PIP in the Leadtek WinFast make it stand out.

Cables and Accessories

Compro made it a point to provide all the required cables with the VideoMate U900 and the VideoMate Action Pro. The Tech-Com SSD-



**Leadtek Winfast
PalmTop TV**
Dual Channel PIP? Wow!